

Assignment 7

TDT4171 — Artificial Intelligence Methods

March 2026

Information

- **Delivery deadline: March 16, 2026 by 23:59.** No late delivery will be graded! Deadline extensions will only be considered for extraordinary situations such as family or health-related circumstances. These circumstances must be documented, e.g., with a doctor's note ("legeerkl ring"). Having a lot of work in other classes is not a legitimate excuse for late delivery.
- Cribbing ("koking") from other students is not accepted, and if detected, will lead to immediate failure of the course. The consequence will apply to both the source and the one cribbing.
- Students can **not** work in groups. Each student can only submit a solution individually.
- Required reading for this assignment: Papers on CBR found on Blackboard "Sources and syllabus".
- For help and questions related to the assignment, **ask the student assistants during the guidance hours.** The timetable for guidance hours can be found under "Assignments" on Blackboard. For other inquiries, an email can be sent to tdt4171@idi.ntnu.no.
- Deliver your solution on Blackboard. Please upload your assignment as a PDF report.

Exercise 1: Core Concepts

- a) Write down the core parts of a case.
- b) Write down the names of the 4R process steps. Describe in your own words each process step.

Exercise 2: Details of a CBR System

- a) Think about an application in which CBR can be applied. Write down the task of the system.
- b) Write down a case representation for your example.
- c) Pick at least one attribute and define a similarity function.

Exercise 3: Knowledge Representation

Draw a table that shows the relationship between the 4R cycle and the knowledge containers.